

The 67th ASH Annual Meeting Abstracts

POSTER

723. ALLOGENEIC TRANSPLANTATION: LONG-TERM FOLLOW-UP, COMPLICATIONS, AND DISEASE RECURRENCE**Vancomycin-resistant enterococci (VRE) colonization is associated with worse outcomes post allogeneic hemopoietic stem cell transplantation**

Haitham Abdelhakim¹, Abdalla Shoaib¹, Marwa Elsayed², Nadia Ghallab³, Leyla Shune¹, Anurag Singh¹, Shaun DeJarnette¹, Anthony Sung¹, Joseph McGuirk¹, Sunil Abhyankar¹

¹ University of Kansas Medical Center, Division of Hematological Malignancy and Cellular Therapeutics, Kansas, United States

² Indiana University, Division of Hematological Malignancy and Cellular Therapeutics, Bloomington, United States

³ University of Vermont, Internal Medicine, Burlington, United States

Abstract Background

Intestinal colonization with vancomycin-resistant enterococci (VRE) is associated with gut microbiome dysbiosis. Furthermore, VRE colonization may help sustain ongoing dysbiosis potentially contributing to persistent microbiota imbalance. Rectal swabs PCR are widely validated as a practical and cost effective method for detecting VRE colonization compared to the gene sequencing needed for microbiota diversity analysis. VRE colonization has been linked to poor outcomes following allo-HSCT but not previously demonstrated in a large patient dataset.

Methods

We reviewed the charts of 759 patients who received an allo-HSCT at the University of Kansas Medical Center between January 2008 and December 2016. A patient was considered VRE-colonized if a rectal swab was positive for VRE either before allo-HSCT or within 30 days after the allo-HSCT. We compared early and late transplant outcomes based on VRE colonization status. We conducted a univariable analysis using Mann-Whitney, chi-square. The Kaplan-Meier and multivariable regression analyses were also performed. Data analyses were conducted using GraphPad Prism and statistical significance was considered at $p < 0.05$.

Results

There were 331 (43.6%) VRE colonized out of total 759 patients included in this study. Patient demographics and transplant-related characteristics were similar between the VRE-colonized and non-colonized groups. The median age was similar (53 vs. 54 years), and male gender was predominant in both groups (59 Vs 61%) for VRE colonized and non-colonized groups respectively. The most common diagnosis was MDS/AML, which accounted for 61.7% of patients in the VRE-colonized group and 55.2% in the non-colonized group. Most patients received peripheral stem cell grafts, with slightly more PSC use among non-colonized patients (80% vs. 73%). Matched unrelated and matched sibling donors were 61.33% and 30.5% in the VRE colonized group vs. 54.6% and 37.6% in the VRE non colonized group. 59% of patients in both groups received reduced-intensity conditioning. GVHD prophylaxis with tacrolimus/MTX was 60% in VRE colonized and 63% in non VRE colonized. Post transplant cyclophosphamide use was limited to haploidentical donors, who numbered 8% in both groups. The median HCT-CI score was 2 in both groups; expected CIBMTR 1-year survival scores were also the same (68%).

There was no difference in the median time to absolute neutrophil count engraftment (16 days) between both groups. However, the platelet recovery was 2 days longer in the VRE-colonized group (19 vs. 17 days, $p < 0.01$). VRE-colonized patients required significantly more transfusion units than non-colonized patients, median packed RBCs units (3 vs 2; $p < 0.001$) and median platelets units (4 vs 2; $p < 0.001$). Patients with VRE colonization had a significantly higher rate of culture-proven bacteremia in the first 30 days post-transplant (22.36% vs 13.08%; $p = 0.0008$). The median hospital stay was significantly longer for VRE-colonized patients than non-colonized patients (26 days vs. 25 days, $p < 0.001$).

Patients with VRE colonization had a significantly higher incidence of grade 2 or higher acute GVHD compared to non-colonized patients (60.73% vs. 49.53%; $p = 0.0026$). They also experienced a higher rate of grade 3 or higher acute GVHD (28% vs. 17.9%; $p = 0.01$). At day 100 post-transplant, VRE-colonized patients demonstrated significantly lower lymphocyte counts than non-colonized controls for both CD4+ T cells (174 vs 204/ μ L; $p = 0.035$) and CD19+ B cells (20 vs 32/ μ L; $p =$

0.036). There was no significant difference observed in 100-day survival between the VRE-colonized (86.6%) and non-colonized (89.1%) groups ($p = 0.29$). However, 1-year survival rates were significantly lower in the VRE-colonized group (58.4%) compared to the non-colonized group 68.1% ($p = 0.0059$). This difference was reflected in the median overall survival, which was 518 days in the VRE-colonized group compared to 1370 days in the non-colonized group ($p < 0.0001$). Additionally, the VRE-colonized group showed an increased cumulative incidence of non-relapse mortality.

Conclusion

While gut microbiota sequencing is limited by cost and turnaround time. VRE colonization is a cost effective and is associated with worse outcomes following allo-HSCT including overall survival.

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